



NATIONAL CHEMISTRY CONTEST - CUBA 2020

"It always seems impossible until it's done."

Nelson Mandela

Name (s): _____ Last name: _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Exam: DAY 2. 10th GRADE

Instructions

- Write all your data clearly in the indicated place (names, surnames, school, municipality, province). Do not type anything in the CODE field.
- Check that your agenda is complete. This is the agenda for the second day of the exam, 10th grade. It is made up of 2 questions (question 6, 17 items; question 7, 5 items) and a total of 5 pages.
- All necessary data, constants, conversions, and fundamental equations are included in the syllabus. You can use a scientific calculator.
- At the beginning of each question its value is indicated (out of a total of 100 points), as well as the value of each item (in marks).

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$

$$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$$

$$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$$

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS																		18
1 H 1.008	2												13	14	15	16	17	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -	
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -	

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 6. Hydrogen compounds (30 points)

6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	6.10
2	6	4	3	2	4	4	2	5	2
6.11	6.12	6.13	6.14	6.15	6.16	6.17		Total	Total
4	2	2	7	3	4	4		60	30

Hydrogen is the lightest and most abundant element in the universe. At standard temperature and pressure, it exists in the molecular form H_2 . This gas is colorless, odorless, insoluble in water and flammable. In the laboratory, hydrogen can be obtained by the reaction between zinc and hydrochloric acid at room temperature, or by heating aluminum or silicon with sodium hydroxide.

6.1. Hydrogen element exists in the form of proton 1H , deuterium 2H and tritium 3H , though other isotopes have been artificially produced. What is the difference between these atoms?

6.2. Formulate the chemical reactions for the generation of hydrogen in the laboratory.

6.3. Compare the distance and bond energy in the hydrogen species H_2 , H_2^+ y H_2^- .

Ion H_3^+ has been detected. It is a highly unstable species, diamagnetic, having a very symmetric structure, in which three atoms are found the closest possible.

6.4. Propose a structure for H_3^+ , showing the special distribution of all atoms; it is not necessary to represent the bonds. How many occupied molecular orbitals present this ion?

Hydrogen forms binary compounds with many elements of the periodic table. These hydrides can be classified according to the chemical bond and the structure that they have in saline (group 1 and 2), metallic (some transition and internal transition elements), intermediate (Cu and Zn) and molecular (Be and groups 13-17).

6.5. Which is the chemical bond predominant in saline hydrides and molecular hydrides?

6.6. Build an approximate energetic diagram of molecular orbital for the ion-molecule of HHe^+ . Why can we predict that this molecule is unstable?

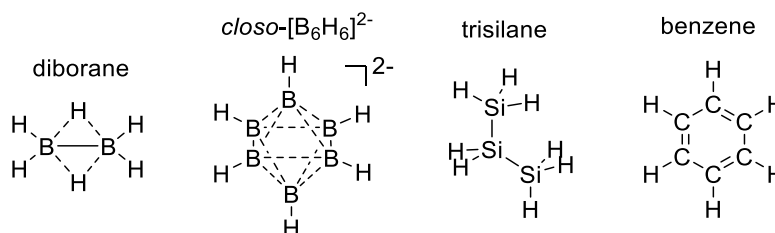
Calcium hydride is used as a portable hydrogen generator in remote places, for example, for filling meteorological balloons. It is convenient because it is cheap and reacts with water at room temperature.

6.7. Write the chemical equation for the obtention of calcium hydride and its reaction with water.

Numerous molecular hydrides are presented in polymeric form. The best examples of this property are the millions of organic molecules that exist.

6.8. How do you explain that carbon forms four chemical bonds in the organic molecules, if the isolated atoms only possess two unpaired electrons?

6.9. Draw the electronic structure of ethyne according to the Valence Bond Theory. Use the hybridization model for the carbon atom.



Boron combines with hydrogen to form boranes, among which the simplest compounds is B₂H₆. In borane molecules the electrons of some hydrogen or boron atoms are shared between three or more atoms. The “unusual” covalent chemical bond is because their molecules are electron deficient, and therefore, cannot complete an octet around boron atoms.

6.10. Why are boranes electron deficient molecules?

6.11. Formulate the chemical equations for the obtention of mixed hydrides LiAlH₄ and NaBH₄ for the corresponding binary hydrides.

The tendency to form polymeric molecules diminishes when going down a group in the periodic table. Carbon is able to virtually form infinite chains in its hydrogen compounds, while the largest analogous silane only contains seven silicon atoms (Si₇H₁₆). On one hand hydrocarbons are stable at room temperature and a heat source is needed to start combustion; on the other hand, SiH₄ spontaneously ignites in the air.

6.12. Mention one factor causing that the silanes are least stable and more reactive than the hydrocarbons.

The boiling point of molecular hydrides for groups 14-17 generally increase while going down the group. The exception to this rule is hydrides H₂O, NH₃ and HF, which possess the highest boiling point of the group they belong to.

6.13. Why H₂O, NH₃ and HF molecules have high boiling points?

6.14. Draw the Lewis' structures for CH₄, NH₃, H₂O and HF. Predict the molecular geometry and the bond angle for the first three of them.

Besides ammonia NH₃, there are numerous binary compounds formed by nitrogen and hydrogen, such as hydrazoic acid HN₃, hydrazine N₂H₄ and diimide or diazene N₂H₂.

6.15. Which is the average oxidation state for nitrogen in the previous molecules?

6.16. Draw resonant structures for azide ion N₃⁻ which is formed when dissolving hydrazoic acid in water. Which is the most contributing resonance structure?

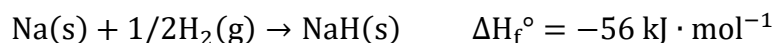
Group 17 hydrides are formed by direct reaction between hydrogen and halogen at high temperatures. There are indirect obtention methods; for example, hydrogen chloride can be obtained by the reaction between solid sodium chloride and concentrated sulfuric acid.

6.17. Formulate the chemical equations corresponding to the direct and indirect obtention methods for hydrogen chloride.

Problem 7. Hydrides in Thermochemistry (20 points)

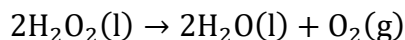
7.1	7.2	7.3	7.4	7.5	Total	Total
7	8	8	8	9	40	20

Sodium hydride is formed by the reaction between hydrogen and sodium.



7.1. Calculate the lattice enthalpy of this substance through the thermodynamic Born-Haber cycle.

Hydrogen peroxide is thermodynamically unstable and slowly decomposes forming water and oxygen. This process is accelerated in the presence of bases.



7.2. Given the values of mean bond dissociation enthalpies, calculate the approximate variation enthalpy for the previous reaction.

Methane is the majority component of natural gas. This hydrocarbon is produced by the putrefaction of organic matter, naturally or because of human action. Its principal application is as fuel.

7.3. Calculate the combustion enthalpy of methane, that is, the enthalpy variation corresponding to the reaction of methane with oxygen.

Part of the methane is found beneath the ocean in the form of combustible ice, which is formed by a clathrate of ideal composition $(\text{CH}_4)_8(\text{H}_2\text{O})_{46}$. These metal hydrides are formed at high pressure and low temperatures.

7.4. Which amount of energy can be obtained for the combustion of 1,00 g of methane hydride? If you cannot solve item 7.3 consider that $\Delta H_{\text{comb}}^\circ(\text{CH}_4) = -900 \text{ kJ} \cdot \text{mol}^{-1}$.

One of the greater dangers because of global warming is that it may cause the release of methane retained in the bottom of the ocean and the permafrost. Methane is a greenhouse gas known with a more potent effect than CO_2 , in consequence, its release in large amounts at the atmosphere would further accelerate the heating effect. It is estimated that the total amount of methane in the earth is larger than $5 \cdot 10^{11}$ tons.

7.5. In how many degrees the earth atmosphere would be heated if that amount of methane is burnt with the atmospheric oxygen? The total heat capacity of the earth atmosphere has an approximate value of $4 \cdot 10^{21} \text{ J} \cdot \text{K}^{-1}$.

$I(\text{H}) = 1312 \text{ kJ/mol}$, $I(\text{Na}) = 496 \text{ kJ/mol}$, $A(\text{H}) = -73 \text{ kJ/mol}$, $A(\text{Na}) = -53 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{H}_2) = 436 \text{ kJ/mol}$, $\Delta H_{\text{dis}}^\circ(\text{Na}_2) = 77 \text{ kJ/mol}$, $\Delta H_{\text{sub}}^\circ(\text{Na}) = 107 \text{ kJ/mol}$, $\Delta H_f^\circ(\text{NaH}) = -56 \text{ kJ/mol}$

Bond	H-O	O-O	O=O
$E_{\text{dis}}/\text{kJ} \cdot \text{mol}^{-1}$	463	146	497

Substance	CH_4	H_2O	CO_2
$\Delta H_f^\circ/\text{kJ} \cdot \text{mol}^{-1}$	-75	-286	-394